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(54) **DOOR LATCH DEVICE AND ACTUATOR**

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CPC **E05B 85/26** (2013.01); **E05B 81/06**
(2013.01)

(58) **Field of Classification Search**

CPC E05B 95/26; E05B 81/06; E05B 81/42;
E05B 81/30

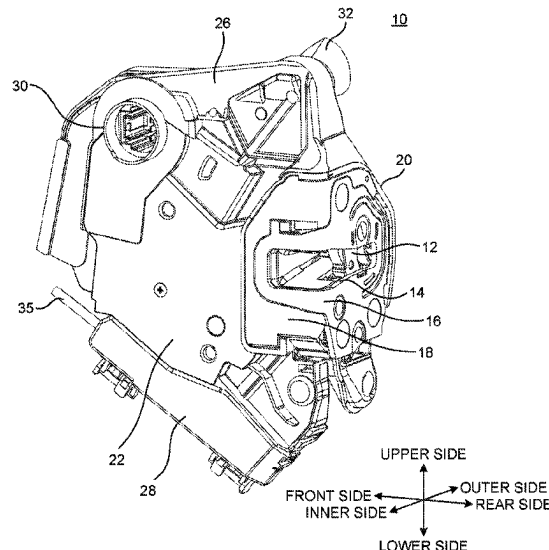
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See application file for complete search history.

(57) **ABSTRACT**

A door latch device includes: a rotating body; first and second levers; a lock mechanism on which an unlocking operation is performed and an open link is disposed in an unlocked position; and a latch mechanism. Further, the lock mechanism includes a lock lever which causes the open link to rotate when being operated by rotation of the rotating body, the rotating body is provided with two tilted walls so as to move the open link to the unlocked position when being rotated in one side from a reference position while the open link is disposed in a locked position, keep the open link in the unlocked position when being rotated in another side to the reference position while the open link is disposed in the unlocked position, and move the open link to the locked position when being further rotated in the another side.

4 Claims, 9 Drawing Sheets



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FIG. 1

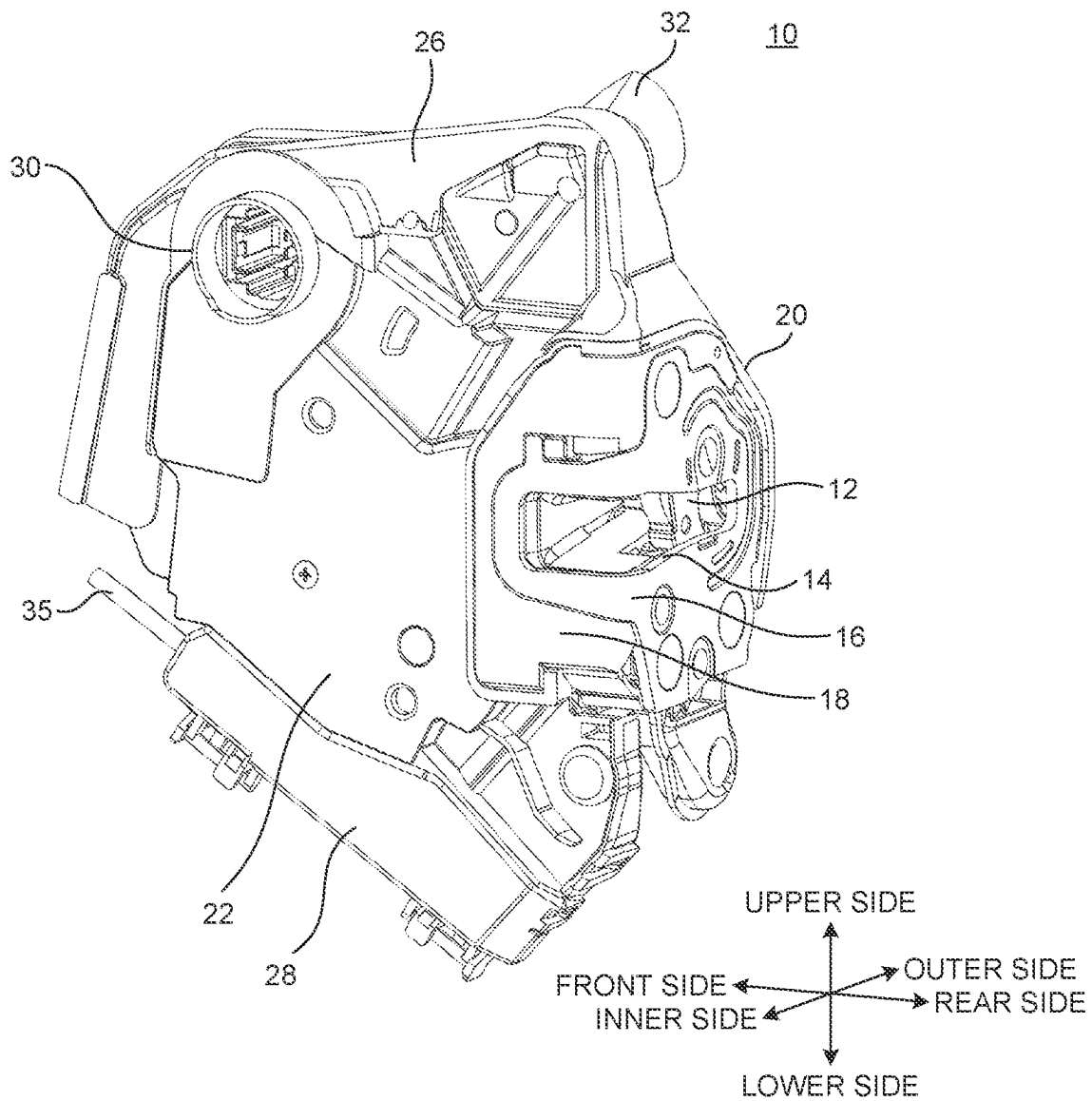


FIG.2

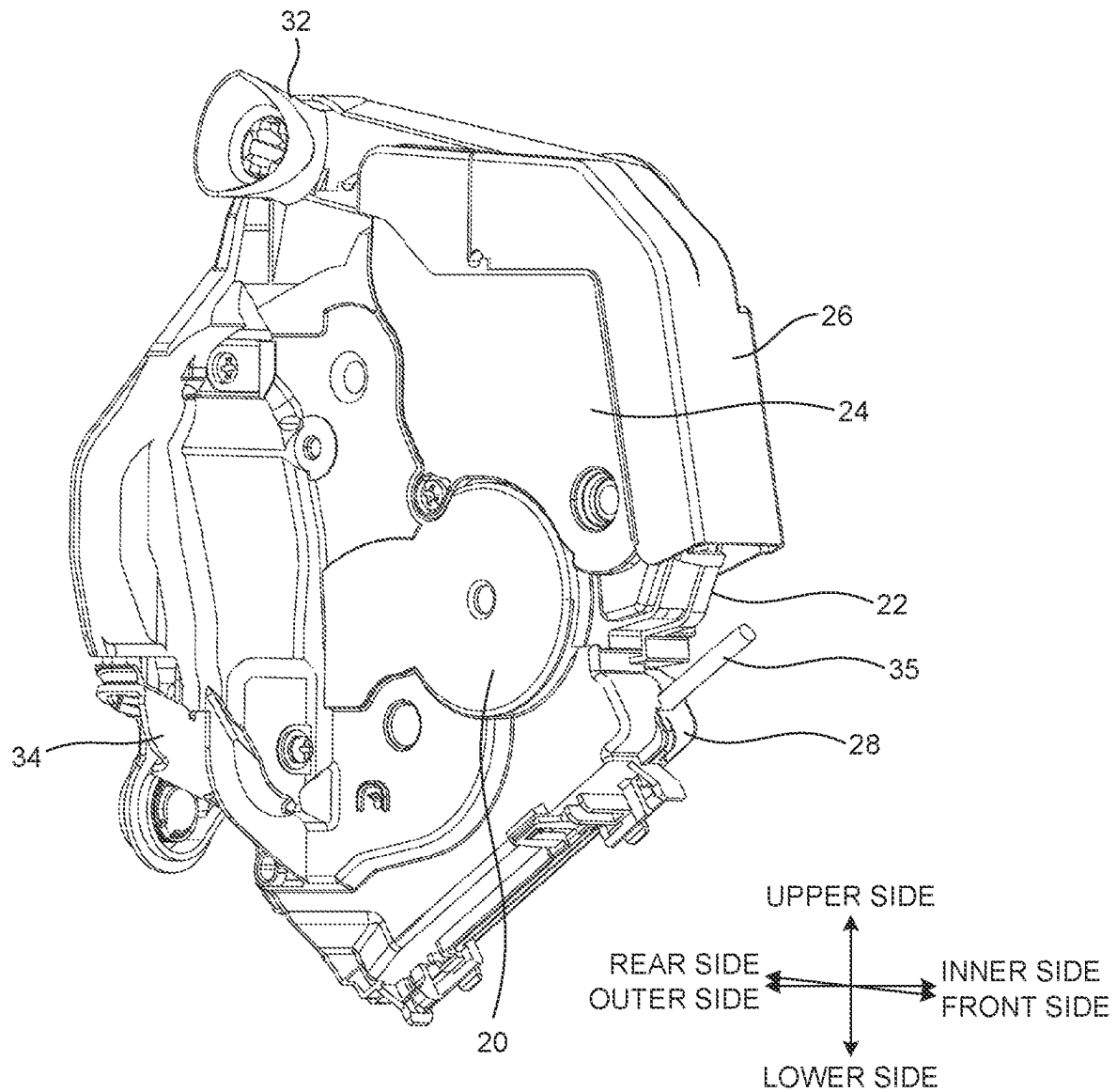
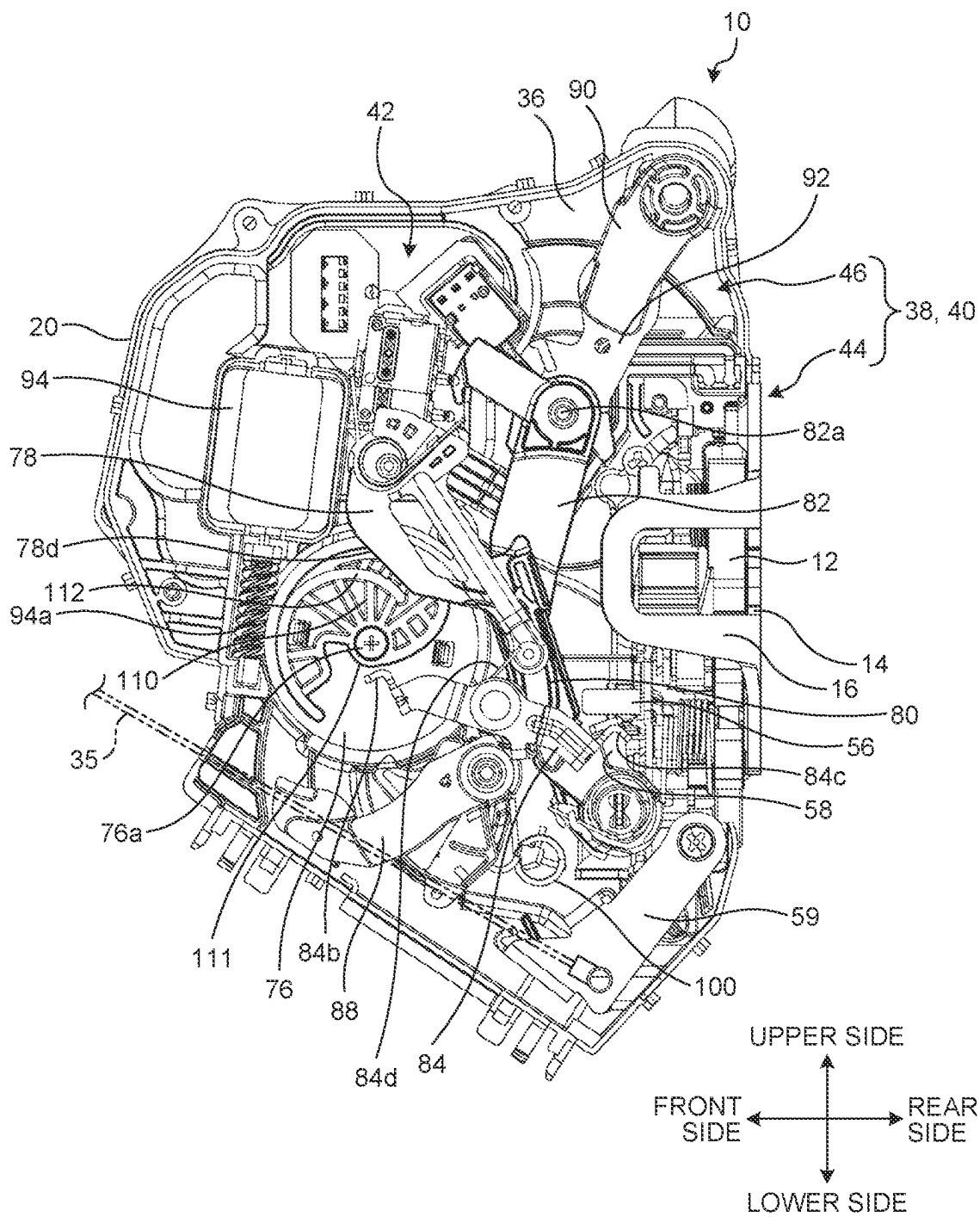


FIG.3



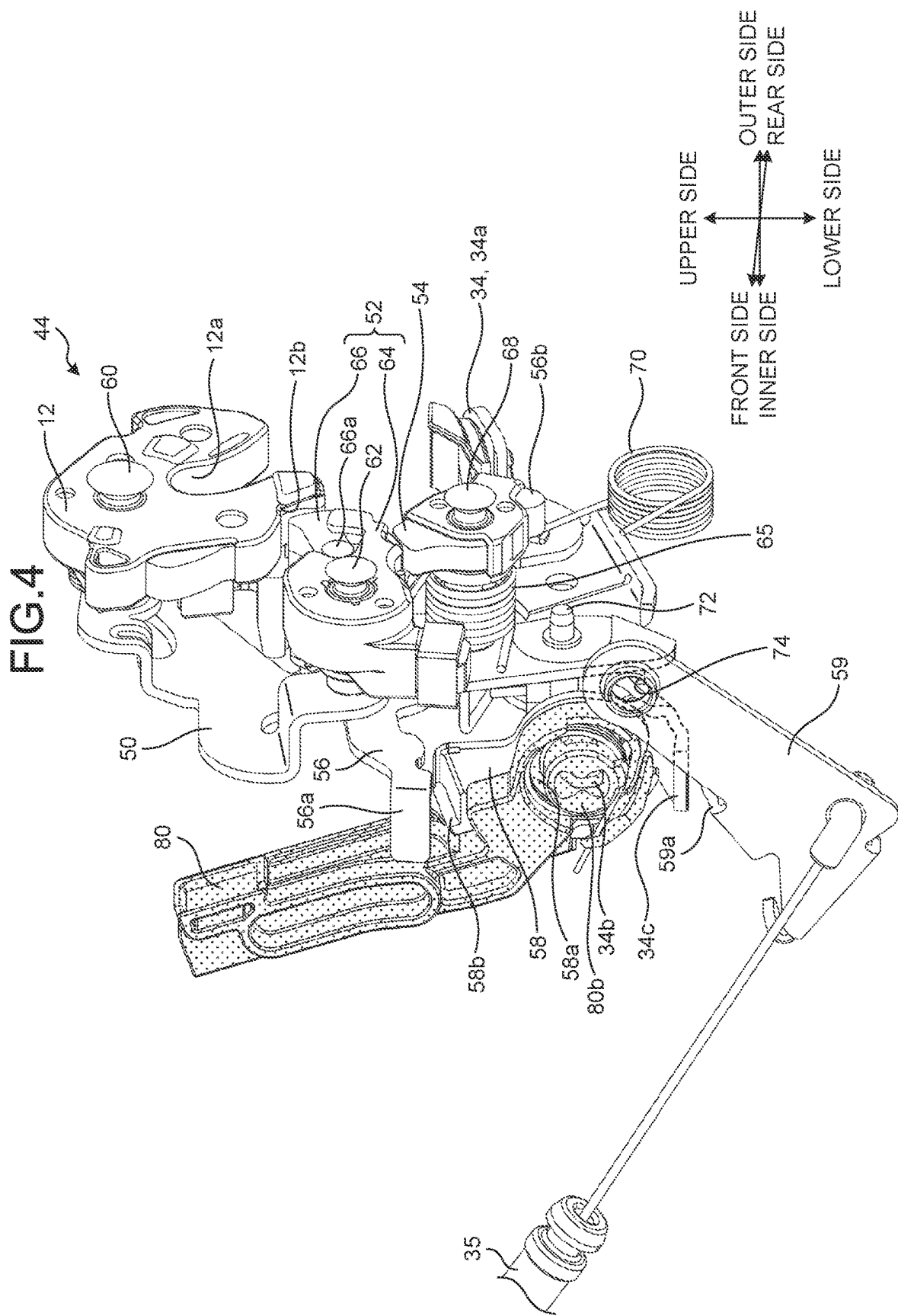


FIG. 5

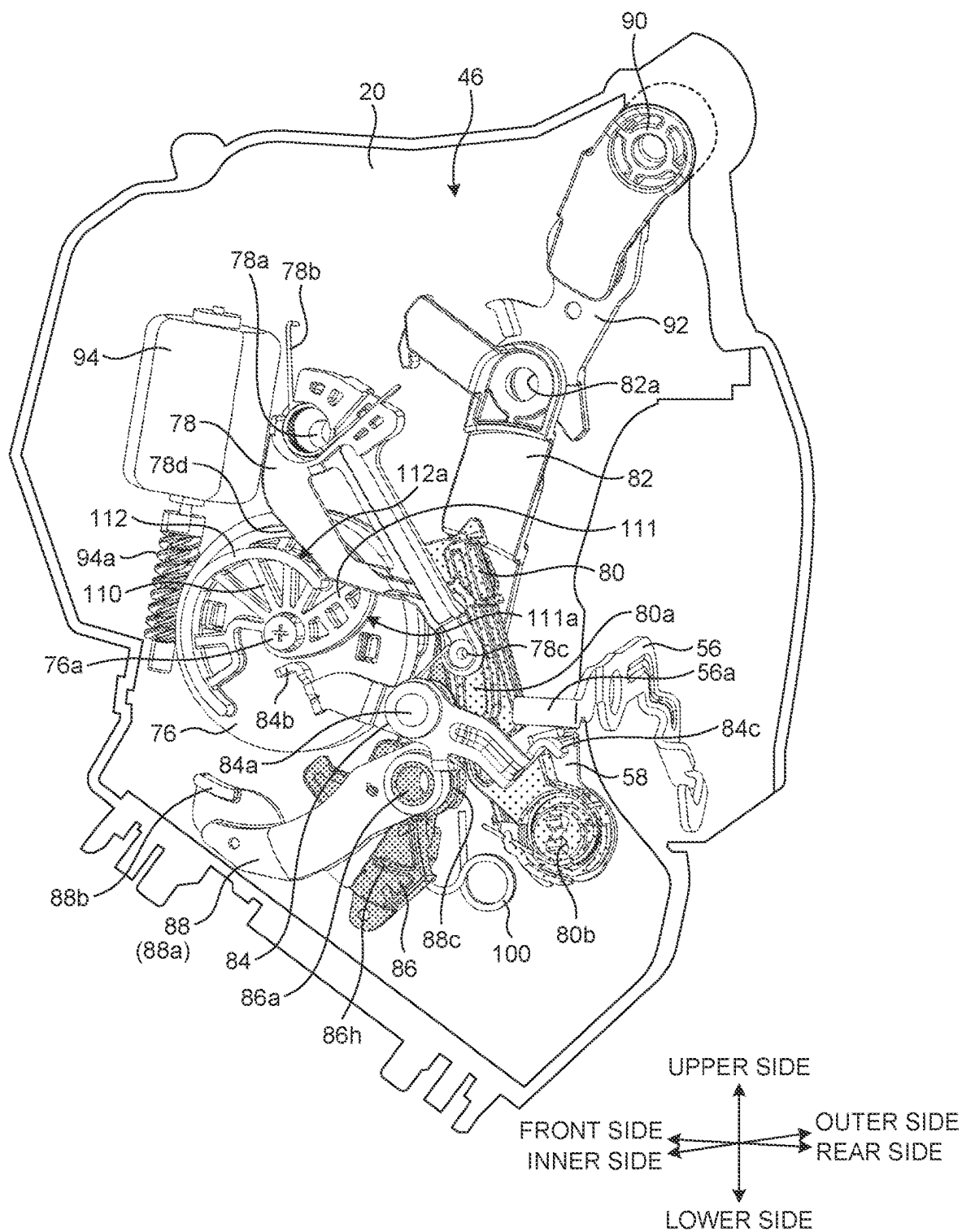


FIG. 6

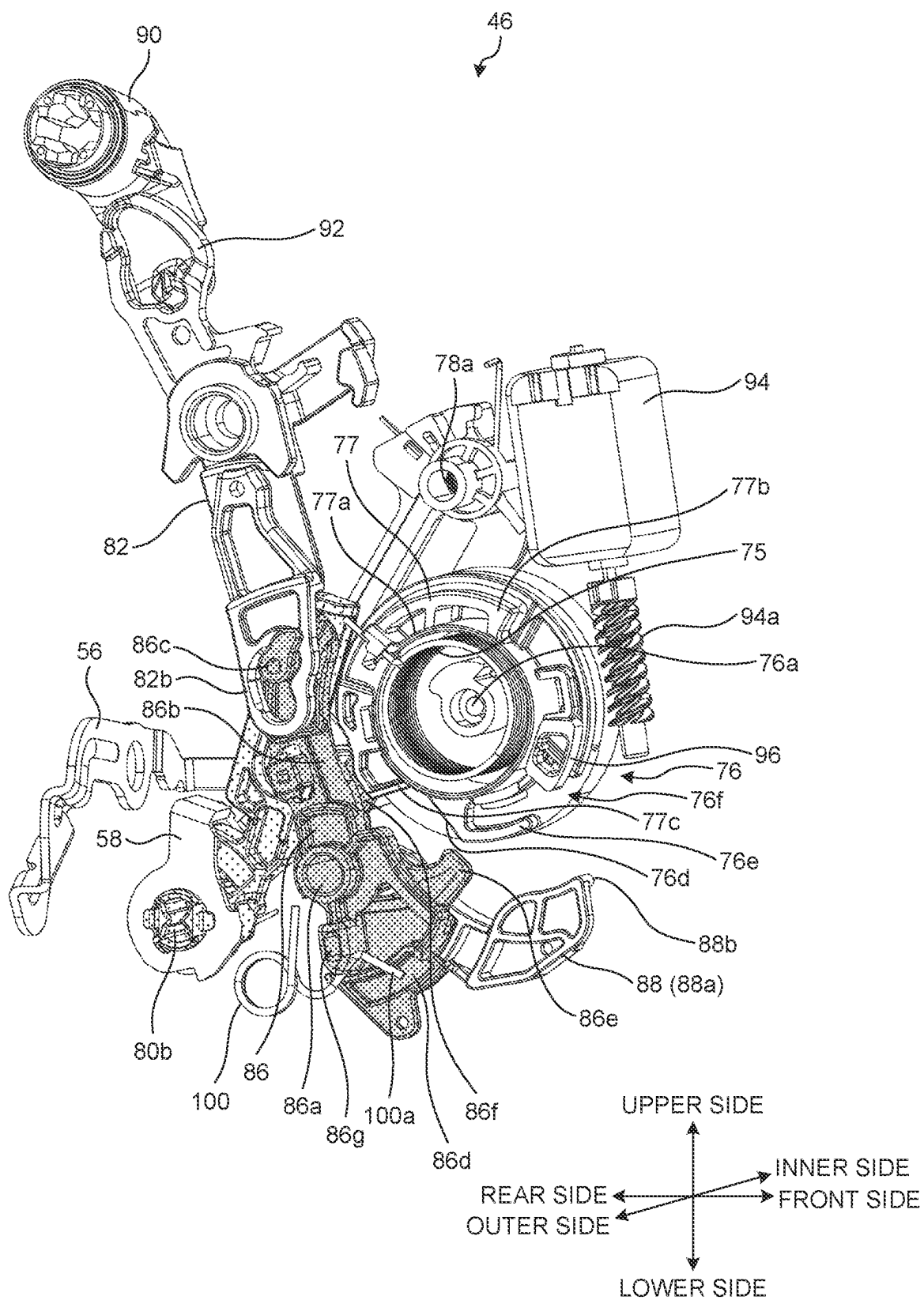


FIG.7A

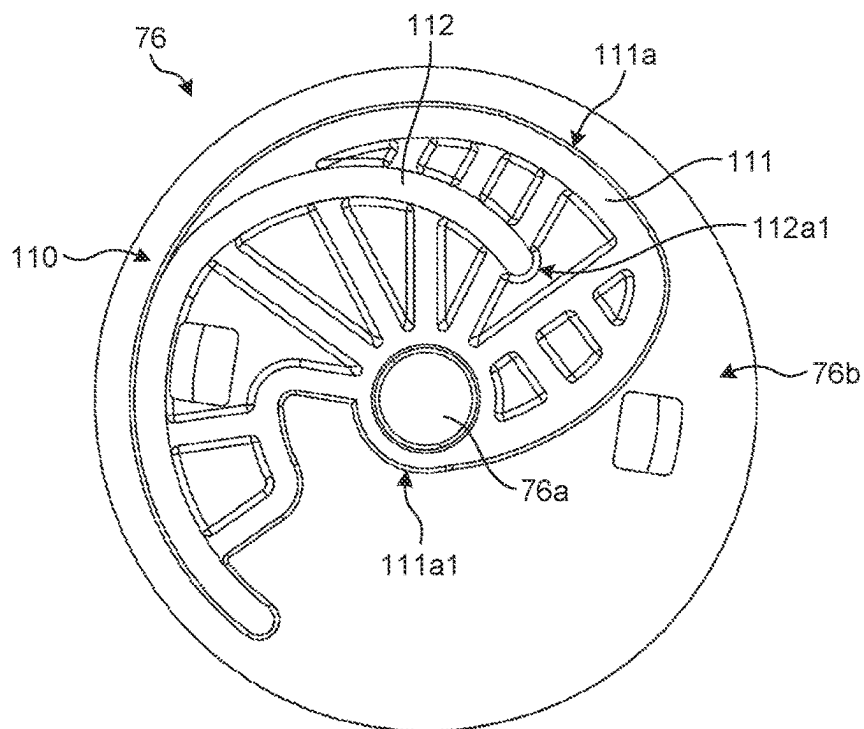
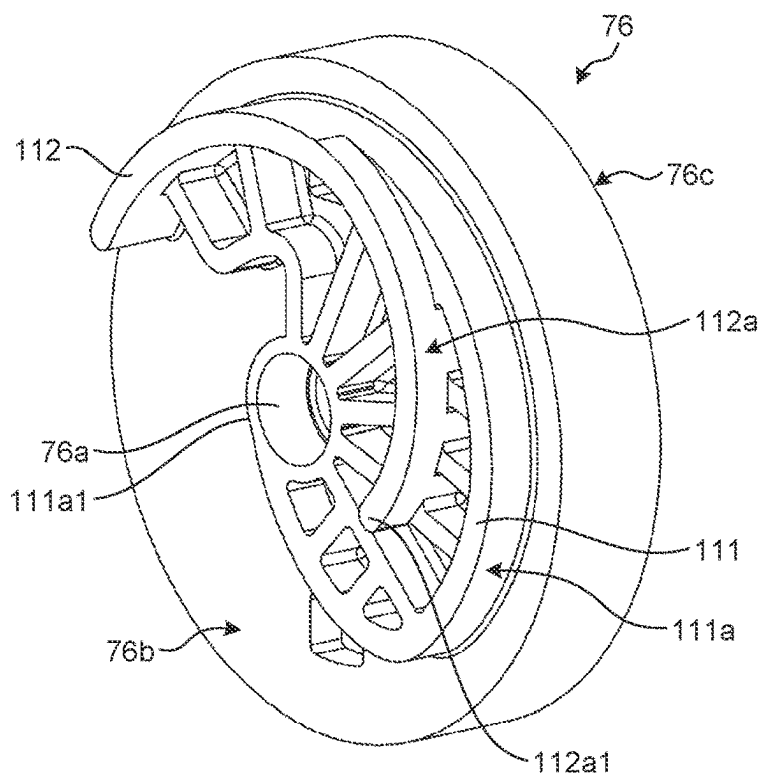


FIG.7B



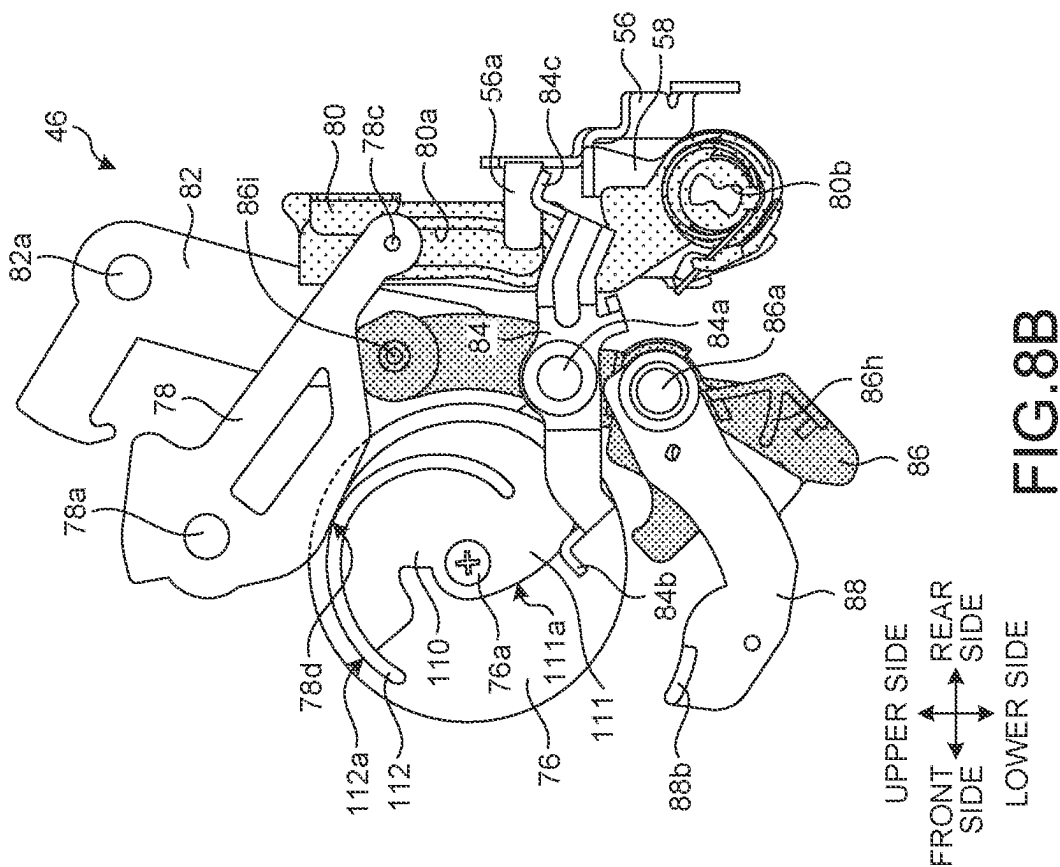


FIG. 8A

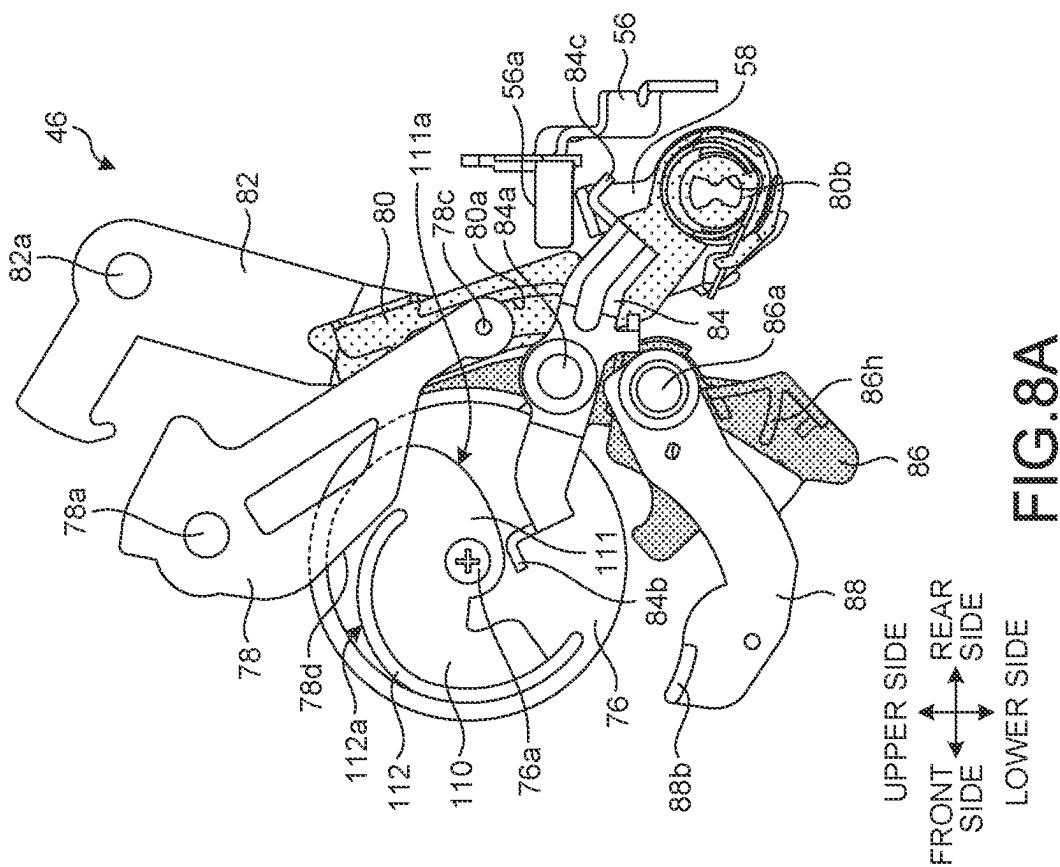


FIG. 8B

FIG.9A

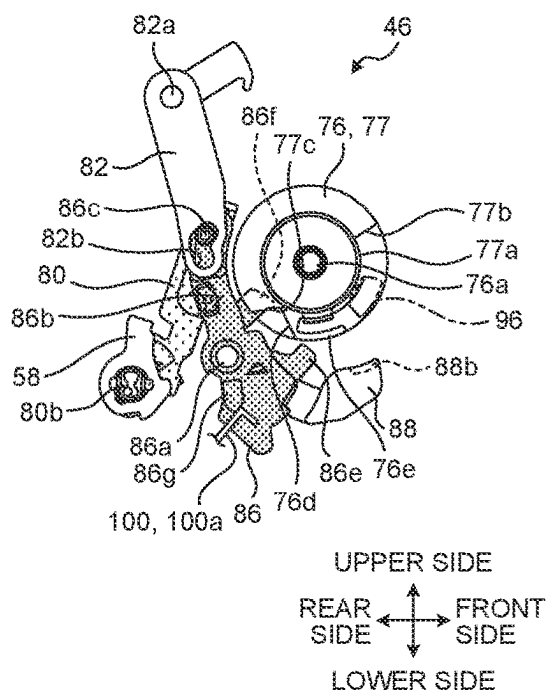


FIG.9B

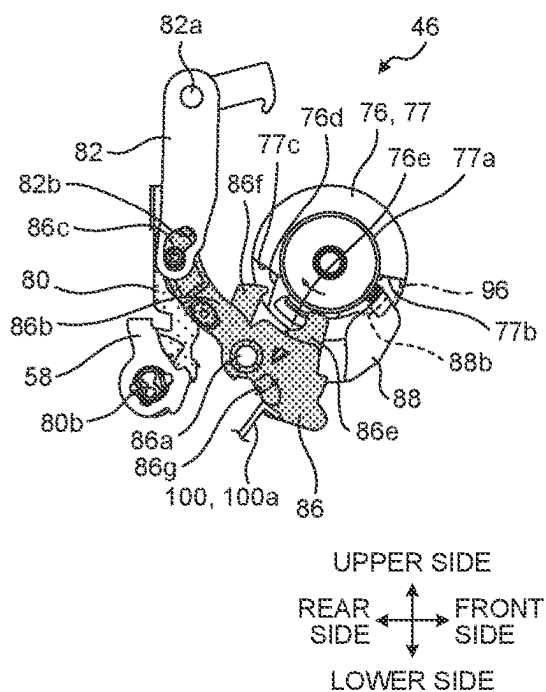


FIG.9C

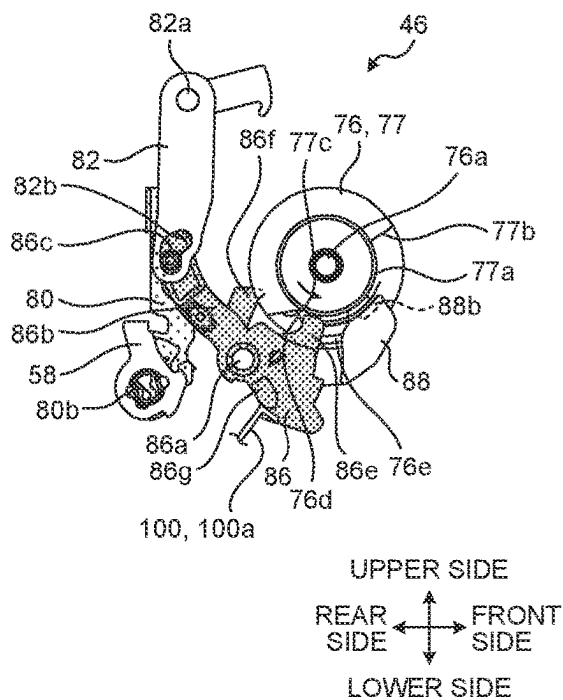
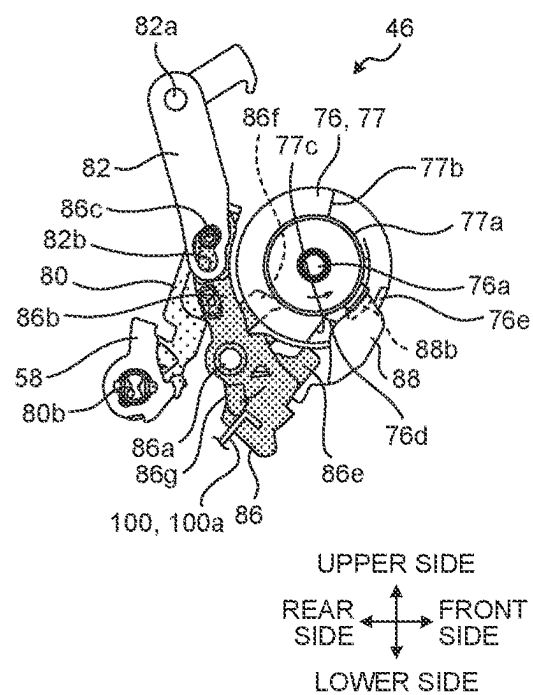


FIG.9D



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DOOR LATCH DEVICE AND ACTUATOR**CROSS-REFERENCE TO RELATED APPLICATION(S)**

This application is a continuation of International Application No. PCT/JP2020/002710, filed on Jan. 27, 2020 which claims the benefit of priority of the prior Japanese Patent Application No. 2019-204785, filed on Nov. 12, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND

The present disclosure relates to a door latch device and an actuator.

A latch mechanism and a lock mechanism are normally disposed on a door latch device mounted on a vehicle. The latch mechanism is configured to maintain a closed state of a door with respect to a vehicle main body by engaging with a striker disposed on the vehicle main body, for example. The lock mechanism is configured to be switched between an unlocked state for releasing an engaging state of the latch mechanism in a case in which an operation of opening the door is performed with a door handle, and a locked state for maintaining the engaging state of the latch mechanism by invalidating the operation for the door handle. As such a type of door latch device, there has been developed a door latch device configured to perform an operation of releasing the engaging state of the latch mechanism and an operation of switching the locked state to the unlocked state by driving of a single actuator. More specifically, a rotating body, is rotated by a motor, the rotating body including a projecting part for causing the latch mechanism to operate disposed on one end face and a projecting part for causing the lock mechanism to operate disposed on an outer peripheral surface, and the projecting parts are respectively caused to work on the latch mechanism and the lock mechanism to successively perform an engagement release operation for the latch mechanism and an unlocking operation for the lock mechanism (for example, refer to Japanese Patent No. 6213927).

SUMMARY

There is a need for providing a door latch device and an actuator by which an increase of the size thereof can be prevented.

According to an embodiment, a door latch device includes: a rotating body to be rotated and driven with respect to a case; a first lever and a second lever disposed to be rotatable with respect to the case, being configured to be operated by a cam part disposed on the rotating body; a lock mechanism on which an unlocking operation is performed so that an open link is arranged at an unlocked position in a case in which the first lever rotates; and a latch mechanism on which a release operation is performed in a case in which the second lever rotates. Further, the cam part is disposed only on one end face of the rotating body, and a first cam to operate the first lever and a second cam to operate the second lever are individually disposed in the cam part, the lock mechanism includes a lock lever to be operated in accordance with rotation of the rotating body to rotate the open link, and the rotating body includes two tilted walls to cause the open link to move to the unlocked position via the lock lever in a case in which the rotating body rotates in one direction from a reference position in a state in which the

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open link is arranged at a locked position, maintain the open link at the unlocked position in a case in which the rotating body rotates in another direction to return to the reference position from a state in which the open link is arranged at the unlocked position, and cause the open link to move to the locked position in a case in which the rotating body further rotates in the another direction from the reference position.

According to an embodiment, an actuator includes: a rotating body configured to be rotated and driven with respect to a case; and a first lever and a second lever that are disposed to be able to rotate with respect to the case, and configured to be operated by a cam part disposed on the rotating body. Further, the cam part is disposed only on one end face of the rotating body, and a first cam configured to operate the first lever and a second cam configured to operate the second lever are individually disposed in the cam part, and a starting end of a cam surface of the second cam is disposed to be closer to an outer peripheral side than a starting end of a cam surface of the first cam is, and an increasing ratio of an outer diameter of the cam surface of the second cam is set to be smaller than an increasing ratio of an outer diameter of the cam surface of the first cam.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a door latch device according to an embodiment of the present disclosure viewed from an outer side and an obliquely rear side of a vehicle;

FIG. 2 is a perspective view of the door latch device viewed from the outer side and an obliquely front side of the vehicle;

FIG. 3 is a side view illustrating an inner part of the door latch device;

FIG. 4 is a perspective view of a latch mechanism;

FIG. 5 is a perspective view of a lock mechanism viewed from an obliquely inner and rear side;

FIG. 6 is a perspective view of the lock mechanism viewed from an obliquely outer and front side;

FIGS. 7A and 7B illustrate a cam wheel,

FIG. 7A is an end face diagram;

FIG. 7B is a perspective view;

FIGS. 8A and 8B are a diagram illustrating an operation of the lock mechanism viewed from an inner side of the vehicle when the cam wheel normally rotates;

FIG. 8A is a diagram illustrating a basic state in which the cam wheel is at a reference position;

FIG. 8B is a diagram illustrating a state in which the cam wheel normally rotates from the reference position by substantially 100 degrees;

FIGS. 9A, 9B, 9C, and 9D are diagrams illustrating the operation of the lock mechanism viewed from the outer side of the vehicle when the cam wheel reversely rotates and normally rotates;

FIG. 9A is a diagram illustrating a basic state in which the cam wheel is at the reference position;

FIG. 9B is a diagram illustrating a state in which the cam wheel reversely rotates from the reference position by substantially 40 degrees;

FIG. 9C is a diagram illustrating a state in which the cam wheel normally rotates by substantially 40 degrees from the state of FIG. 9B; and

FIG. 9D is a diagram illustrating a state in which the cam wheel normally rotates by substantially 40 degrees from the state of FIG. 9C.

DETAILED DESCRIPTION

With the door latch device in the related art, the release operation for the latch mechanism and the unlocking opera-

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tion for the lock mechanism can be performed by a single motor, so that there are advantages such that the number of components can be reduced, for example. However, components such as a lever to be engaged with the projecting part to be operated are required to be disposed in a region 5 opposed to the end face of the rotating body and a region opposed to the outer peripheral surface of the rotating body, so that the size of the device may be increased.

The following describes a preferred embodiment of a door latch device according to the present disclosure in detail 10 with reference to the attached drawings.

FIG. 1 to FIG. 3 are diagrams illustrating the door latch device according to the embodiment of the present disclosure. Although not illustrated in the drawings, the door latch device exemplified herein is mounted on a side door disposed 15 on the right side of a four-wheeled automobile, and performs opening/closing control for the side door by changing an engaging state with a striker disposed on a vehicle main body in accordance with an opening/closing operation with a door handle or a locking/unlocking operation with a 20 key.

As illustrated in FIG. 1 and FIG. 2, in a door latch device 10, a latch member 12 for latching the striker is disposed in the rear of a striker entry groove 14. The latch member 12 is part of a latch mechanism 44 (described later). The striker 25 entry groove 14 is formed as part of a cover plate 16. A body 18 is disposed around the cover plate 16. A vehicle inner side and a vehicle rear side of the latch mechanism 44 are covered by the cover plate 16 and the body 18.

The door latch device 10 is covered by a case 20, a first 30 cover 22, and a second cover 24 in addition to the cover plate 16 and the body 18. The case 20 mainly covers a vehicle outer side, the first cover 22 mainly covers the vehicle inner side, and the second cover 24 further covers a front upper part on the vehicle inner side of the case 20. The cover plate 35 16, the body 18, the case 20, the first cover 22, and the second cover 24 constitute a housing of the door latch device 10.

The door latch device 10 further includes a waterproof cover 26 that covers an upper part, a cable cover 28 on a 40 lower part on the vehicle inner side, a coupler 30 disposed on an upper part on the vehicle inner side, and a key cylinder coupling part 32 disposed on an upper part on the vehicle outer side. The waterproof cover 26 covers a boundary part between the case 20 and the first cover 22, and prevents 45 infiltration of water. The cable cover 28 covers a connecting portion for a cable 35. The cable 35 links the door latch device 10 with an inner handle (not illustrated). The coupler 30 is connected to a harness connector (not illustrated). The key cylinder coupling part 32 is a portion to be operated 50 when a key is inserted therein. On a portion positioned on the vehicle outer side of the door latch device 10, an end part of art outer lever 34 connected to an outer handle (not illustrated) is disposed to project toward the outside.

As illustrated in FIG. 3, an accommodation space 36 is 55 formed on the vehicle inner side of the door latch device 10. The accommodation space 36 is a region in which the vehicle outer side is covered by the case 20, and the vehicle inner side is mainly covered by the first cover 22. The vehicle inner side of the accommodation space 36 is also 60 covered by the cover plate 16, the body 18, and the cable cover 28 in addition to the first cover 22.

The accommodation space 36 briefly includes a mechanism region 40 in which the machine mechanism 38 is disposed, and an electric component region 42 in which 65 electric components are disposed. The electric component region 42 occupies an upper part on the vehicle front side,

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and the mechanism region 40 occupies a residual portion. The machine mechanism 38 houses the latch mechanism 44 that latches and unlatches the striker by the latch member 12, and a lock mechanism. 46 that is switched between a locked state and an unlocked state. The latch mechanism 44 is 5 disposed on the vehicle rear side in the accommodation space 36, and covered by the cover plate 16 and the body 18.

The machine mechanism 38 houses an electric release unit that can release, by power of a motor 94, a latched state 10 of the striker by the latch mechanism 44, and a manual release unit that can release, by manual operation, the latched state of the striker by the latch mechanism 44. The electric release unit includes the motor 94, a cam wheel (rotating body) 76, and the like (described later) to unlatch the striker. The manual release unit unlatches the striker via 15 an inner lever 59 (described later) and the outer lever 34 mechanically linked with a manual operation.

As illustrated in FIG. 4, the latch mechanism 44 includes a base bracket 50, a ratchet 52, a ratchet holder 54, a ratchet 25 lever 56, an anti-panic lever 58, and the inner lever 59 in addition to the latch member 12 and the outer lever 34 described above. The respective elements of the latch mechanism 44 are supported by the base bracket 50.

The latch member 12 is supported by the housing via a latch shaft 60 in a rotatable manner, and includes a striker engagement groove 12a and a ratchet engagement part 12b. 30 When the striker enters the striker engagement groove 12a in accordance with a door-closing operation from a door-opened state, the latch member 12 rotates against elastic force of a spring (not illustrated), and the ratchet 52 engages with the ratchet engagement part 12b to latch the striker at a full-latch position.

The ratchet 52 includes a base lever 64 supported by the housing via a ratchet shaft 62 in a rotatable manner, and a pole lever 66 supported by the base lever 64 via a base shaft 35 part 66a in a rotatable manner. The base lever 64 is elastically energized by a base spring 65. The pole lever 66 is bent in a predetermined angle range with respect to the base lever 64. A substantially straight attitude or the ratchet 52 is held by being supported by the ratchet holder 54 from a lateral side, and a distal end of the pole lever 66 engages with the ratchet engagement part 12b to hold the latch member 12 at the full latch position.

The ratchet holder 54 is supported by the housing via a shaft part 68 in a rotatable manner, and elastically energized 45 by a holder spring 70 to support a lateral side of the base lever 64. The ratchet holder 54 rotates against elastic force of the holder spring 70 when the ratchet lever 56 rotates, and is separated from the base lever 64. When the ratchet holder 54 is separated from the base lever 64, the base lever 64 and the pole lever 66 of the ratchet 52 are caused to be in a buckling state centered on the base shaft part 66a, and the pole lever 66 leaves the ratchet engagement part 12b to open the latch member 12. As a result, the latch member 12 is 50 rotated by elastic force of a spring (not illustrated), and the striker is unlatched. In a case of operating the ratchet 52 via the ratchet holder 54, the operation can be performed with smaller force as compared with a case of directly operating the ratchet 52.

The ratchet lever 56 is supported by the base bracket 50 in a rotatable manner, and includes a passive part 56a projecting toward the vehicle inner side with respect to a rotation axis and a rotation working part 56b projecting toward the vehicle outer side with respect to the rotation 60 axis. In the ratchet lever 56, the rotation working part 56b causes the ratchet holder 54 to rotate when the passive part 56a moves toward the upper side.

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The outer lever **34** is supported by the housing via a shaft part **72** in a rotatable manner, and includes a handle operation part **34a** projecting toward the vehicle outer side with respect to the shaft part **72**, and a working part **34b** and a lever passive piece **34c** projecting toward the vehicle inner side with respect to the shaft part **72**. The handle operation part **34a** is a portion operated by the outer handle. The working part **34b** is inserted into a hole **58a** of the anti-panic lever **58** and an odd-form hole **80b** of an open link **80** (described later). The lever passive piece **34c** is disposed on a lower side of the working part **34b**, and operated by the inner lever **59**. The outer lever **34** is rotated by an operation of the handle operation part **34a** or the lever passive piece **34c**, and pushes up the anti-panic lever **58**.

The inner lever **59** is supported by the housing via the shaft part **74** in a rotatable manner, and is rotated by an operation of the cable **35**. Accordingly, an operation piece **59a** pushes up the lever passive piece **34c**.

The anti-panic lever **58** includes the hole **58a** into which the working part **34b** is inserted, and a working piece **56b** that is bent at an upper part. The anti-panic lever **58** is pushed up by the working part **34b** when the open link **80** (described later) is at an unlocked position and the outer lever **34** rotates, and the working piece **56b** pushes up the passive part **56a** of the ratchet lever **56**. Accordingly, the ratchet holder **54** and the ratchet **52** perform an unlatch operation. The anti-panic lever **58** has a structure separated from the open link **80** due to an anti-panic mechanism.

As illustrated in FIG. 5 and FIG. 6, the lock mechanism **46** includes a cam wheel **76** supported by the housing via a shaft part **76a** in a rotatable manner, a knob lever (first lever) **78** that is supported by the housing via a shaft part **78a** in a rotatable manner and driven by the cam wheel **76**, the open link **80** that is driven by the knob lever **78**, a sub-lock lever **82** linked with the open link **80**, and an open lever (second lever) **84** that is supported by the housing via a shaft part **84a** in a rotatable manner and driven by the cam wheel **76**. The lock mechanism **46** further includes a lock lever **86** and an auxiliary lever **88** linked with the sub-lock lever **82**, and a key lever **90** and a sub-key lever **92** that are linked with a key operation and drive the sub-lock lever **82**. To easily identify components in the respective drawings, the lock lever **86** is represented by fine dots, and the open link **80** is represented by coarse dots.

The cam wheel **76** has a disk shape having teeth (not illustrated) on an outer peripheral surface, meshes with a worm **94a** disposed on a drive shaft of the motor **94** via the teeth on the outer peripheral surface, and can rotate in a case in which the motor **94** is driven. In the following description, for convenience' sake, a case in which the cam wheel **76** rotates clockwise in FIG. 5 is assumed to be normal rotation, and a case in which the cam wheel **76** rotates counterclockwise is assumed to be reverse rotation.

A cam part **110** is disposed on the cam wheel **76**. The cam part **110** is disposed only on an end face **76b** positioned on the vehicle inner side of the cam wheel **76** (hereinafter, simply referred to as an inner end face **76b**), and projects toward the vehicle inner side. A first cam **111** and a second cam **112** are disposed on the cam part **110**. Cam surfaces **111** and **112a** having different shapes are disposed on respective outer circumferences of the first cam **111** and the second cam **112**. Each of the cam surfaces **111a** and **112a** is formed so that a distance from the center of the cam wheel **76** is increased when the cam wheel **76** rotates counterclockwise.

More specifically, as illustrated in FIG. 5 and FIGS. 7A and 7B, regarding the cam surface **111a** of the first cam **111** positioned on a side close to the inner end face **76b** of the

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cam wheel **76** in the cam part **110**, a minimum outer diameter portion close to a center part of the cam wheel **76** is assumed to be a starting end **111a1**, and the cam surface **111a** extends in a direction of rotating counterclockwise in FIGS. 7A and 7B. In the example illustrated in FIGS. 7A and 7B, the cam surface **111a** of the first cam **111** is disposed in a range of substantially 270 degrees from the starting end **111a1**. As is clear from FIGS. 7A and 7B, the cam surface **111a** of the first cam **111** is configured to have a distance from the center that is gradually increased in a range of substantially 100 degrees from the starting end **111a1**, and to be a circular arc having a fixed outer diameter after the range of 100 degrees. A maximum outer diameter of the cam surface **111a** of the first cam **111** is set at a distance with which the open lever **84** (described later) can rotate counterclockwise to the maximum in FIG. 5 to perform an unlatch operation on the ratchet holder **54** and the ratchet **52** via the ratchet lever **56**.

On the other hand, regarding the cam surface **112a** of the second cam **112** positioned on a side separated from the inner end face **76b** of the cam wheel **76** in the cam part **110**, a position rotated counterclockwise from the starting end **111a1** of the first cam **111** by substantially 120 degrees is assumed to be a starting end **112a1**, and the cam surface **112a** extends in a direction of rotating counterclockwise in FIGS. 7A and 7B. In the example illustrated in FIGS. 7A and 7B, the cam surface **112a** of the second cam **112** is configured to have a distance from the center that is gradually increased in a range of substantially 100 degrees from the starting end **112a1**, and to be a circular arc having a fixed outer diameter that is the same as that of the cam surface **111a** of the first cam all after the range of 100 degrees. A maximum outer diameter of the cam surface **112a** of the second cam **112** is set at a distance with which the knob lever **78** (described later) can rotate counterclockwise to the maximum in FIG. 5 and the open link **80** is caused to have an erect attitude.

The starting end **112a1** of the cam surface **112a** of the second cam **112** is disposed to be closer to an outer peripheral side than the starting end **111a1** of the cam surface **111a** of the first cam **111** is, and an increasing ratio of the outer diameter of the cam surface **112a** of the second cam **112** is smaller than an increasing ratio of the outer diameter of the cam surface **111a** of the first cam **111**.

As illustrated in FIG. 6, the cam wheel **76** includes an auxiliary component **77** on an end face **76c** positioned on the vehicle outer side (hereinafter, simply referred to as an outer end face **76c**). The auxiliary component **77** is integrally attached to the cam wheel **76** in a state of being prevented from relatively rotating. A torsion coil spring (elastic member) **75** is disposed inside a cylindrical part **77a** formed by the auxiliary component **77**. The torsion coil spring **75** is configured to have a cylindrical coil part by winding a metal fine wire, one spring end part is engaged with the cam wheel **76**, and the other spring end part is engaged with the case **20**. The torsion coil spring **75** allows the cam wheel **76** to normally rotate and reversely rotate with respect to the case **20** in a case in which external force is applied to the cam wheel **76**, while energizing the cam wheel **76** to be maintained at a neutral reference position in a case in which external force is removed.

The auxiliary component **77** includes a regulating projection **77b** projecting toward the vehicle inner side in the vicinity of an outer circumference, and a first tilted wall **77c** that is disposed on a substantially opposite side of the regulating projection **77b**. The regulating projection **77b** abuts on an elastic member **96a** of a rotation stopper **96**

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disposed in the case 20 to regulate rotation of the cam wheel 76 when the cam wheel 76 reversely rotates. The first tilted wall 77c is tilted from an outer peripheral surface of the cylindrical part 77a toward an outer peripheral side to be gradually positioned on a clockwise direction side in FIG. 6. The first tilted wall 77c is disposed to secure a gap between itself and the outer end face 76c of the cam wheel 76.

The cam wheel 76 further includes a second tilted wall 76d and a holding wall 76e. The second tilted wall 76d is tilted from the outer peripheral surface of the cylindrical part 77a of the auxiliary component 77 toward the outer peripheral side to be gradually positioned on a counterclockwise direction side in FIG. 6. The first tilted wall 77c and the second tilted wall 76d extend to be gradually separated from each other from a position close to the cylindrical part 77a toward the outer peripheral side. The first tilted wall 77c of the auxiliary component 77 is disposed to be closer to the vehicle outer side than the second tilted wall 76d is. The holding wall 76e is a circular arc wall extending from a portion positioned on the outer peripheral side of the second tilted wall 76d toward the counterclockwise direction in FIG. 6. As illustrated in FIG. 6, an end part on the clockwise direction side of the holding wall 76e is closed by a portion constituting the second tilted wall 76d, while an end part on the counterclockwise direction side thereof is adjacent to a notch 76f opening on the outer peripheral surface of the cam wheel 76.

As illustrated in FIG. 5, a follower surface 78d that abuts on the cam surface 112a of the second cam 112 in the cam part 110 is disposed on the knob lever 78. The follower surface 78d is driven by the second cam 112 when the cam wheel 76 normally rotates, and functions to cause the knob lever 78 to rotate counterclockwise against a lever spring 78b. A knob 78c is disposed at a distal end part of the knob lever 78. The knob 78c engages with a side surface guide groove 80a disposed on the open link 80. The knob lever 78 functions such that, when being rotated counterclockwise in FIG. 5, the knob 78c moves toward the upper side along the side surface guide groove 80a to erect the tilted open link 80, and when being rotated clockwise in FIG. 5, the knob 78c moves toward a lower side along the side surface guide groove 80a to cause the erect open link 80 to be tilted toward the vehicle front side.

The open link 80 includes the odd-form hole 80b at a lower end, and rotates about the lower end as a center to be switched between an erect attitude (unlocked position) and a tilted attitude (locked position) as described above. The lock mechanism 46 is caused to be in the locked state in a case in which the open link 80 is arranged at the locked position illustrated in FIG. 5, and the lock mechanism 46 is caused to be in the unlocked state in a case in which the open link 80 is arranged at the unlocked position. That is, when the open link 80 is at the locked position, the anti-panic lever 58 is tilted together with the open link 80, so that the anti-panic lever 58 does not abut on the ratchet lever 56 even in a case of being lifted up by the outer lever 34. Thus, the ratchet lever 56 does not operate, and the door is kept closed in the locked state. On the other hand, when the open link 80 is at the unlocked position, the anti-panic lever 58 erects together with the open link 80, so that the anti-panic lever 58 abuts on and pushes up the ratchet lever 56 when being lifted up by the outer lever 34. Accordingly, the ratchet lever 56 operates, and the unlocked state in which the door can be opened is caused.

The working part 34b of the outer lever 34 is inserted into the odd-form hole 80b of the open link 80. In a case in which the outer lever 34 operates, the open link 80 moves along an

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upper and lower direction. The anti-panic lever 58 is assembled to the lower end part of the open link 80. The anti-panic lever 58 moves integrally with the open link 80.

The sub-lock lever 82 is supported by the housing via a shaft part 82a in a rotatable manner, engaged with the lock lever 86 via an outer knob 86c of the lock lever 86, and engaged with the open link 80 arranged at the locked position via an inner knob 86i. That is, when the sub-lock lever 82 rotates counterclockwise in FIG. 5 due to rotation of the key lever 90 and the sub-key lever 92, the lock lever 86 rotates clockwise in FIG. 5 via the outer knob 86c, and the lock lever 86 is pushed out by the inner knob 86i to cause the open link 80 to be at the unlocked position. In a case in which the sub-lock lever 82 rotates clockwise and returns to an original position, the open link 80 is returned to the locked position by the knob lever 78 that rotates clockwise in FIG. 5 by elastic force of the lever spring 78b.

The open lever 84 is a constituent element of the electric release unit. That is, the open lever 84 operates to open the door in a case in which the motor 94 is driven by a switch operation and the like performed by a driver, and includes a cam follower part 84b extending from the shaft part 84a toward the vehicle front side and a ratchet operation part 84c extending toward the vehicle rear side. An open spring 84d is disposed between the open lever 84 and the housing. The open spring 84d energizes the cam follower part 84b in the clockwise direction in FIG. 5 to abut on the cam surface 111a of the first cam 111 in the cam part 110 of the cam wheel 76. In a case in which the cam wheel 76 normally rotates in FIG. 5, the first cam 111 pushes down the cam follower part 84b, the open lever 84 rotates counterclockwise about the shaft part 84a against the open spring 84d, and the ratchet operation part 84c moves upward. When the ratchet operation part 84c moves upward, the passive part 56a of the ratchet lever 56 is pushed up, the latch mechanism 44 is unlatched, and the door is opened. When the cam wheel 76 returns to the neutral reference position, the open lever 84 also returns to an original attitude by the open spring 84d.

The open lever 84 can operate the ratchet lever 56 independently of the open link 80. Thus, with the door latch device 10 including the open lever 84, the door can be opened by the electric release unit even when the lock mechanism 46 is in the locked state (that is, the open link 80 is at the locked position).

As illustrated in FIG. 6, the lock lever 86 includes an arm 86b that is supported by the housing in a rotatable manner via a shaft part 86a and extends toward the upper side, the outer knob 86c projecting from a distal end of the arm 86b toward the vehicle outer side, a downward extending part 86d extending toward the lower side, a first projection 86e projecting from the downward extending part 86d toward the vehicle front side, a second projection 86f projecting from the vicinity of the shaft part 86a of the arm 86b toward the vehicle front side, a spring reception part 86g projecting from the downward extending part 86d toward the vehicle outer side, and two push-out parts 86h projecting from a surface on the vehicle inner side of the downward extending part 86d toward the inner side of the vehicle. The outer knob 86c is engaged with the sub-lock lever 82 by being fit into a guide hole 82b formed at the lower end of the sub-lock lever 82. The lock lever 86 is rotated via the outer knob 86c when the sub-lock lever 82 rotates, and can be displaced between a working position for switching the open link 80 from the locked position to the unlocked position, and a non-working position not working on the open link 80.

The spring reception part **86g** abuts on a bent part **100a** of a lock spring **100**. The lock spring **100** defines an attitude of the sub-lock lever **82** via the spring reception part **86g**.

The first projection **86e** abuts on the first tilted wall **77c** in a case in which the cam wheel **76** rotates counterclockwise in FIG. 6. When the cam wheel **76** continuously rotates counterclockwise in a state in which the first tilted wall **77c** abuts on the first projection **86e**, the lock lever **86** rotates clockwise in FIG. 6. The second projection **86f** abuts on the second tilted wall **76d** in a case in which the cam wheel **76** rotates clockwise in FIG. 6. When the cam wheel **76** continuously rotates clockwise in a state in which the second tilted wall **76d** abuts on the second projection **86f**, the lock lever **86** rotates counterclockwise in FIG. 6. The second projection **86f** is disposed to be closer to the vehicle inner side than the first tilted wall **77c** is. The second projection **86f** enters a gap between the first tilted wall **77c** and the outer end face **76c** of the cam wheel **76**, and does not abut on the first tilted wall **77c**. The two push-out parts **86h** abut on the auxiliary lever **88** in a case in which the lock lever **86** rotates clockwise in FIG. 5.

As illustrated in FIG. 5, the auxiliary lever **88** is supported by the shaft part **86a** of the lock lever **86** in a rotatable manner, and includes an arm **88a** projecting from the shaft part **86a** toward the vehicle front side, and a circular arc projection **88b** disposed on an upper part of a distal end of the arm **88a**. The circular arc projection **88b** abuts on the holding wall **76e** of the cam wheel **76** in a case of rotating counterclockwise in FIG. 6. An auxiliary lever spring **88c** is interposed between the auxiliary lever **88** and the lock lever **86**, the auxiliary lever spring **88c** energizing the auxiliary lever **88** in a direction of rotating counterclockwise in FIG. 5 with respect to the lock lever **86**. A lower surface of the auxiliary lever **88** is kept abutting on the push-cut parts **86h** due to elastic force of the auxiliary lever spring **88c**.

FIGS. 8A and 8B are diagrams illustrating an operation of the lock mechanism **46** viewed from the vehicle inner side when the cam wheel **76** normally rotates (rotates clockwise in FIGS. 8A and 8B). The following describes the operation of the lock mechanism **46** with reference to these drawings as appropriate.

As illustrated in FIG. 8A, in a basic state in which the cam wheel **76** is at the reference position, the cam surface **112a** of the second cam **112** is separated from the follower surface **78d** of the knob lever **78**, and the cam follower part **84b** is separated from the cam surface **111a** of the first cam **111** at a position where the open lever **84** rotates clockwise to the maximum. At this point, the open link **80** is arranged at the locked position, and the knob **78c** of the knob lever **78** is positioned on a lower side of the side surface guide groove **80a**.

When the cam wheel **76** starts to normally rotate by being driven by the motor **94** from the basic state, a portion in the vicinity of the starting end **112a1** of the cam surface **112a** of the second cam **112** abuts on the follower surface **78d** of the knob lever **78**, and the knob lever **78** starts to rotate counterclockwise in FIGS. 8A and 8B. Substantially at the same time, a portion in the vicinity of the starting end **111a1** of the cam surface **111a** of the first cam **111** abuts on the cam follower part **84b** of the open lever **84**, and the open lever **84** starts to rotate counterclockwise in FIGS. 8A and 8B.

As described above, the increasing ratio of the outer diameter of the cam surface **111a** of the first cam **111** is larger than that of the cam surface **112a** of the second cam **112**. Thus, in a case in which the cam wheel **76** normally rotates, as illustrated in FIG. 8B, the passive part **56a** of the ratchet lever **56** is sufficiently pushed up by the open lever

84 at an early stage, the latch mechanism **44** unlatches the striker, and the door is opened. On the other hand, the knob lever **78** is caused to be in a state of being rotated counterclockwise to the maximum by the cam surface of the second cam **112**, the open link **80** gradually rotates clockwise when the knob **78c** moves toward the upper side along the side surface guide groove **80a**, and the open link **80** is caused to be at the unlocked position. The sub-lock lever **82**, the lock lever **86**, and the auxiliary lever **88** do not operate in this period, and are maintained in the basic state in FIG. 8A.

During the operation described above, the torsion coil spring **75** interposed between the case **20** and the cam wheel **76** is gradually bent. Thus, when electric supply to the motor **94** is stopped thereafter, the cam wheel **76** rotates counterclockwise to be at the reference position due to elastic force of the torsion coil spring **75**, and the lock mechanism **46** returns to the basic state illustrated in FIG. 8A.

At the time of such electric release, as illustrated in FIG. 8A and FIG. 8B, the open lever **84** can be rotated by being driven by the motor **94** to work on the latch mechanism **44** to unlatch the striker. At this point, the open link **80** also reciprocates between the locked position and the unlocked position. The open link **80** does not work on the other components, but synchronously operates at appropriate time intervals at the time of electric release, so that it is possible to prevent grease from being hardened due to long-term deterioration, or prevent a spring, a lever, and the like made of steel material from getting rusty. Accordingly, a smooth operation of the lock mechanism **46** can be secured at all times.

At the time of electric release, only the open link **80** synchronously operates, and the lock lever **86** does not operate. Thus, the spring reception part **86g** of the lock lever **86** does not get over the bent part **100a**, and sound is not generated, so that a user does not feel a sense of incongruity.

FIGS. 9A to 9D are diagrams illustrating an operation of the lock mechanism **46** viewed from the vehicle outer side when the cam wheel **76** reversely rotates (rotates clockwise in FIGS. 9A to 9D) and normally rotates (rotates counterclockwise in FIGS. 9A to 9D). The following further describes the operation of the lock mechanism **46** with reference to the drawings as appropriate.

As illustrated in FIG. 9A, in the basic state in which the cam wheel **76** is arranged at the reference position, the lock lever **86** is at a position where the lock lever **86** rotates clockwise to the maximum, and the first projection **86e** and the second projection **86f** are in a non-abutting state with respect to the cam wheel **76**.

When the cam wheel **76** starts to reversely rotate by being driven by the motor **94** from the reference state, and rotation of the cam wheel **76** proceeds to substantially 40 degrees as illustrated in FIG. 9B, the second tilted wall **76d** of the cam wheel **76** abuts on the second projection **86f**. Accordingly, the lock lever **86** rotates counterclockwise, and the spring reception part **86g** is displaced to get over the bent part **100a** of the lock spring **100**.

When the lock lever **86** rotates, the sub-lock lever **82** is rotated clockwise via the outer knob **86c**, the open link **80** is rotated counterclockwise via the inner knob **86i**, and the auxiliary lever **88** is rotated counterclockwise by the push-out parts **86h**. Accordingly, each of the sub-lock lever **82** and the open link **80** is caused to be at the unlocked position, and the circular arc projection **88b** of the auxiliary lever **86** is displaced to a position close to the cylindrical part **77a** via the notch **76f**.

When the spring reception part **86g** gets over the bent part **100a** or the lock spring **100**, the cam wheel **76** starts to

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normally rotate by being driven by the motor **94**. As illustrated in FIG. **9C**, when normal rotation of the cam wheel **76** proceeds to substantially 40 degrees from the state of FIG. **9B**, the cam wheel **76** returns to the reference position illustrated in FIG. **9A**. However, the spring reception part **86g** is held by the bent part **100a**, so that the lock lever **86**, the sub-lock lever **82**, and the open link **80** are kept in the attitude illustrated in FIG. **9B**. Accordingly, the lock mechanism **46** is caused to be in the unlocked state. At this point, due to normal rotation of the cam wheel **76**, the holding wall **76e** is arranged on an outer peripheral side of the circular arc projection **88b**, and the circular arc projection **88b** is caused to be in a state of engaging with an inner peripheral surface of the holding wall **76e**.

When normal rotation of the cam wheel **76** further proceeds by substantially 40 degrees from the state illustrated in FIG. **9C**, the first tilted wall **77c** of the cam wheel **76** abuts on the first projection **86e**. Accordingly, the lock lever **86** rotates clockwise, and as illustrated in FIG. **9D**, the spring reception part **86g** gets over the bent part **100a** of the lock spring **100** to return to the reference state illustrated in FIG. **9A**.

When the lock lever **86** rotates clockwise, the sub-lock lever **82** is rotated counterclockwise via the outer knob **86c**, the open link **80** is rotated clockwise by the knob lever **78** that is rotated by elastic force of the lever spring **78b**, and each of the sub-lock lever **82** and the open link **80** returns to the reference state illustrated in FIG. **9A**.

On the other hand, the circular arc projection **88b** engages with the holding wall **76e** of the cam wheel **76**, so that the auxiliary lever **88** is maintained in a state of rotating counterclockwise against elastic force of the auxiliary lever spring **88c**. When normal rotation of the cam wheel **76** further proceeds from this state, the regulating projection **77b** abuts on the rotation stopper **96** of the case **20** via the elastic member **96a**, and rotation of the cam wheel **76** is stopped. Due to this, excessive rotation of the cam wheel **76** can be prevented.

After the regulating projection **77b** abuts on the rotation stopper **96** and rotation of the cam wheel **76** is stopped, the cam wheel **76** starts to reversely rotate by being driven by the motor **94**. When reverse rotation of the cam wheel **76** proceeds to the position illustrated in FIG. **9A**, engagement between circular arc projection. **88b** and the holding wall **76e** is released. Due to this, the auxiliary lever **88** is rotated clockwise by elastic force of the auxiliary lever spring **88c**, and returns to the position illustrated in FIG. **9A**. In this way, the entire lock mechanism **46** returns to the basic state illustrated in FIG. **9A**.

In this way, with the door latch device **10** described above, the unlatch operation for the latch mechanism **44** and switching of the lock mechanism **46** between the locked state and the unlocked state can be performed by the single motor **94**. Furthermore the cam part **110** is disposed only on the one inner end face **76b** of the cam wheel **76**, so that the knob lever **78** and the open lever may be disposed only in a region opposed to the inner end face **76b** of the cam wheel **76**, and the levers **78** and **84** are not disposed in a region opposed to the outer peripheral surface of the cam wheel **76**. Thus, it possible to prevent a situation in which the size of the door latch device **10** is increased.

The individual cam surfaces **111a** and **112a** are caused to abut on the knob lever **78** and the open lever **84**, so that the levers **78** and **84** can be operated at a desired optional timing without influencing mutual operations by appropriately

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changing profiles of the cam surfaces **111a** and **112a**, and the door latch device **10** satisfying needs of the user can be applied.

In the embodiment described above, the first cam and the second cam are disposed on the outer peripheral surface of the cam part, but the embodiment is not limited thereto. Any one of the cam surfaces or both of the cam surfaces may be formed on the inner peripheral surface of the cam part.

In the embodiment described above, exemplified is the actuator applied to the door latch device, but the actuator can also be applied as an actuator for another device.

According to the present disclosure, the cam part is disposed only on the one end face of the rotating body, so that each of the first lever and the second lever may be disposed only in a region opposed to the one end face of the rotating body, and each lever is not disposed in a region opposed to the outer peripheral surface of the rotating body. Accordingly, it is possible to prevent the size of the device from being increased.

Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

What is claimed is:

1. A door latch device comprising:

a latch mechanism which maintains a door closed relative to a vehicle body by engaging with a striker to bring the striker into a latched state;

a motor;

a rotating body configured to be rotated and driven by the motor with respect to a case;

and a lock mechanism including i) a first lever and a second lever which are disposed to be rotatable with respect to the case and are configured to be operated by a cam part disposed on the rotating body and ii) an open link which moves between a locked position and an unlocked position,

wherein the lock mechanism switches between i) a locked state in which the open link is at the locked position such that a releasing operation of the latched state by a manual operation is inactivated and ii) an unlocked state in which the open link is at the unlocked position such that the releasing operation of the latched state by the manual operation is activated,

wherein the open link is moved to the unlocked position when the first lever is rotated by the cam part,

wherein the latched state of the striker is released when the second lever is rotated by the cam part when the lock mechanism is in the locked state,

wherein the cam part is disposed only on one end face of the rotating body,

wherein the cam part includes a first cam configured to operate the first lever and a second cam configured to operate the second lever which are individually disposed in the cam part,

wherein when the state of the latch mechanism is set to an unlatched state, not only the second lever but also the first lever are rotated so as to rotate the open link, thereby increasing a frequency of stirring grease inside the case by the open link,

wherein the first lever includes a knob disposed at a distal end part thereof,

wherein the open link includes a side surface guide groove, and

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wherein the knob of the first lever engages with the side surface guide groove of the open link so that the open link rotates when the knob moves along the side surface guide groove.

2. The door latch device according to claim 1, wherein 5
each of the first cam and the second cam has a cam surface on an outer circumference, and

the cam surface of the first cam and the cam surface of the second cam are disposed side by side with each other in an axial direction of the rotating body. 10

3. The door latch device according to claim 2, wherein the cam surface of the first cam and the cam surface of the second cam are configured to partially match with each other.

4. The door latch device according to claim 1, wherein an 15
elastic member, which is configured to maintain the rotating body at a neutral position with respect to the case, is disposed between the case and another end face of the rotating body.

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